

Proposal Title:

Machine Learning-Based Classification of Benign and Malignant Breast Lesions Using Radiomic Features from MRI Data (T1 DCE, T2W, and ADC Scans)

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Background and Objective:

Breast cancer diagnosis often relies on radiological evaluation of MRI scans to determine whether a biopsy is required. Radiologists assess tumoral lesions in both breasts and assign BIRADS (Breast Imaging-Reporting and Data System) scores, which guide the selection of lesion sites for biopsy. The primary objective of this project is to leverage radiomic features extracted from MRI scans to train supervised machine learning models that classify breast lesions as benign or malignant.

Methodology:**1. Data Acquisition**

- ☐ MRI scans of breast tissue from patients.
- ☐ Scan types include:
 - T1-weighted Dynamic Contrast Enhanced (T1 DCE), shows tumor vascularity and contrast uptake.
 - T2-weighted (T2W) – shows tissue water content and edema.
 - Apparent Diffusion Coefficient (ADC), derived from diffusion-weighted imaging, reflects tissue cellularity.
- ☐ Clinical labels: BI-RADS scores assigned by radiologists; patient metadata, if relevant.

2. Lesion Segmentation:

- Two experienced radiologists segment lesions independently to evaluate intra- and inter-observer variability.

3. Radiomic Feature Extraction:

- Features have been extracted from pre-registered MRI scans (T1 DCE, T2W, and ADC).

4. Labeling:

- Lesions are labeled according to BIRADS scores:

- BIRADS 1, 2, 3 → Benign
- BIRADS 4, 5 → Malignant

5. Machine Learning Classification:

- Supervised ML models will be trained on the extracted radiomic features to classify lesions.
- For each scan type (T1 DCE, T2W, ADC), there are two independent segmentation masks per image, resulting in **six datasets** (3 scan types $\times$ 2 masks).
- Models will be run separately on each dataset to evaluate inter-observer variability.
- Additional analyses will be conducted using concatenated datasets:
 - **Pairwise concatenation:** Combining two scan types at a time (T1+T2, T1+ADC, T2+ADC).
 - **Triple concatenation:** Combining all three scan types (T1+T2+ADC).
- **Post-hoc explainability:** After model training, SHAP and LIME will be applied to interpret predictions and identify the most influential radiomic features for lesion classification.
- **Cross-validation strategy:** Because the dataset is small (<200 samples), appropriate validation techniques will be used, such as **nested cross-validation**, **leave-one-out cross-validation (LOOCV)**, or **k-fold cross-validation**. Nested CV is particularly well-suited for small medical imaging datasets and will help prevent overfitting while providing reliable performance estimates.

5. Outputs and Performance Metrics:

For each model and dataset, the following outputs will be saved and reported:

- Confusion matrix
- ROC-AUC
- Accuracy
- Precision and recall
- Cohen's kappa value
- Calibration curve
- Negative Predictive Value (NPV) and Positive Predictive Value (PPV)
- p-value correlation with Benjamini-Hochberg FDR correction

Team Composition and Roles:

- **Physicians (2):** Data collection and patient information.
- **Radiologists (2):** Independent lesion segmentation, interpretation, and annotation of medical data.
- **PhD Student:** Data handling, extraction of radiomic features, and manuscript preparation.
- **Bachelor's/MSc Student:** Development and training of ML models on datasets derived from both segmentation masks.
- **Supervisors (3 Assistant Professors):** Oversight and guidance throughout the project.

Expected Outcomes:

- ☐ Reliable segmentation masks with quantified inter-observer variability.
- ☐ Comprehensive radiomic feature datasets for each scan type and mask combination.
- ☐ Machine learning models capable of distinguishing benign from malignant lesions with high accuracy.
- ☐ Post-hoc explainable predictions highlighting key radiomic features influencing classification (SHAP and RADAR plots).
- ☐ Complete performance evaluation including traditional metrics, calibration, predictive values, and statistically corrected correlations.